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03/09/2023 PHYSICS

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Q1. Contract.

Q2. Repulsion.

Q3. Virtual, erect and laterally inverted.

Q4. Below the image of the fish that can be seen by the hunter.

Q5. $P = \frac{1}{f}$

$$\Rightarrow 1.5 = \frac{1}{f}$$

$$f = \frac{1}{1.5}$$

$$f = \frac{10}{15} = \frac{2}{3}$$

$$f = +0.667\text{m} \quad (\text{approx.})$$

The lens prescribed is a converging lens
i.e., it is convex lens because the
focal length is positive.

Teacher's Signature

Q6.

$$I = 5A$$

$$t = 30 \text{ min} = 30 \times 60 \text{ seconds}$$

$$t = 1800 \text{ sec}$$

$$I = \frac{q}{t}$$

$\Rightarrow$

$$q = It$$

$$q = 5 \times 1800$$

$$q = 9000 \text{ coulombs.}$$

Q7.

Bulb - 

Resistor / Resistance - 

Switch / Key (Open) -  OR 

Switch / Key (Closed) -  OR 

Battery - 

Ammeter - +(A) -

In

Q8: As it is a convex lens, so focal length will be positive.

$$f = +12 \text{ cm}$$

$$u = -20 \text{ cm}$$

$$h_o = 5 \text{ cm}$$

My Lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{12} = \frac{1}{v} - \left(-\frac{1}{20} \right)$$

$$\frac{1}{12} = \frac{1}{v} + \frac{1}{20}$$

$$\frac{1}{12} - \frac{1}{20} = \frac{1}{v}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{5-3}{60}$$

$$\frac{1}{v} = \frac{2}{60}$$

$$\frac{1}{v} = \frac{1}{30}$$

$$\Rightarrow v = +30 \text{ cm}$$

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Answer $v = 30 \text{ cm}$

Q9.

In lens, $m = +\frac{v}{u} = \frac{h_i}{h_o}$

$$\frac{h_i}{5} = \frac{30}{-20} = \frac{-3}{2}$$

$$h_i = -1.5 \times 5 = -7.5 \text{ cm}$$

So, the position = 30 cm behind the lens.

nature = Real, Inverted and magnified.

Size = -7.5 cm below the principal axis.

Q9.

$$V_1 = 10 \text{ V}$$

$$V_2 = 5 \text{ V}$$

$$W = 5 \text{ J}$$

$$Q = ?$$

So, V between the ends of two conductors = $V_1 - V_2$

$$V = 10 - 5 = 5 \text{ V}$$

$$\text{Now, } V = \frac{W}{Q}$$

$$\Rightarrow Q = \frac{W}{V}$$

$$Q = \frac{5}{5} = 1 \text{ Coulomb.}$$

So, the amount of charge flowing between the two ends is 1C.

Q16: a) Pole: (denoted by P)

Pole is the centre of the concave mirror and is a point on the contact of the mirror and

Principal Axis ~~where the light get reflected~~

b) Radius of Curvature: (denoted by C)

Radius of Curvature is the radius of the hollow glass sphere from which the concave mirror is made from. It is double the focal length.

c) Principal Axis:

It is an imaginary line passing through the pole and centre of curvature and is perpendicular to the surface of the mirror is called Principal Axis.

d) Principal Focus: (denoted by F)

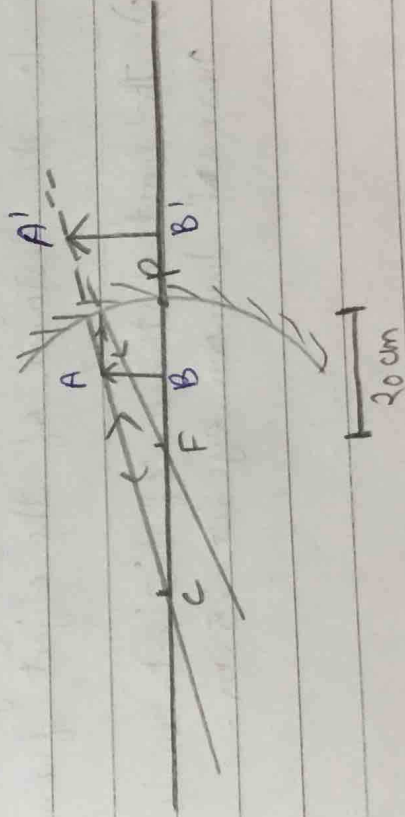
The point on the principal axis where all the rays parallel to principal axis ~~approach~~ converge after reflection is called principal Focus.

$f = -20 \text{ cm}$

The range of distance of the candle flame from the mirror such that we can obtain a erect image of the candle flame using a concave mirror is $0 \text{ cm} - 20 \text{ cm}$.

The two characteristics of the image are:

- 1) Virtual image
- 2) Magnified image.



$AB = \text{height of candle flame}$

$A'B' = \text{height of Image.}$

03/09/2022

Q11. a) Sun appears to be reddish in the morning because during sunrise sun is near the horizon. Light travelling to the observer travels a larger distance. So most of the blue light and shorter wavelength rays are scattered away by air. Hence light reaching the observers eye is of longer wavelength i.e., red, orange etc. and the sun appears reddish.

b) Power of accommodation of eye is the ability or power of the eye to change and alter the focal length and size of eye lens according to the distance of the object from the eye.

c) The least distance of distinct vision for a normal human eye is 25 cm.

Q1. The

It

Q2. Am

Q3. Both of

~~Q4.~~ A

Q5. An

20

0

0

2

A

S

A

S